



Pediatrics Examination

Question paper

Exam code: 143YVH2 | Date: 2026-10-07 | 50 questions | 50 minutes

Name: _____

Training level / Section: _____ Score: _____

Instructions: Choose the single best answer for each question. Mark A, B, C, or D on the answer sheet. Each question carries one point.

1. The upper-to-lower body segment ratio is approximately what value at birth, reflecting the relatively large head and trunk?

- A. About 1.7
- B. About 1.0
- C. About 0.7
- D. About 2.5

2. A toilet-trained girl voids normally at regular intervals yet is constantly damp with urine day and night. Which anomaly best explains this?

- A. Primary nocturnal enuresis without daytime symptoms
- B. An ectopic ureter inserting below the urinary sphincter
- C. An overactive bladder producing urgency incontinence
- D. A small simple bladder wall diverticulum

3. In girls with central precocious puberty, which underlying cause is most common?

- A. A hypothalamic hamartoma compressing the median eminence
- B. A pineal germinoma secreting human chorionic gonadotropin
- C. Idiopathic activation of the hypothalamic-pituitary-gonadal axis
- D. Cranial irradiation injuring the hypothalamic region

4. Which set names the four core principles of biomedical ethics?

- A. Confidentiality, consent, capacity, and competence
- B. Veracity, fidelity, advocacy, and stewardship
- C. Autonomy, beneficence, nonmaleficence, and justice
- D. Privacy, dignity, equity, and solidarity

5. A 7-year-old develops an acute eruption of numerous small, drop-like, scaly papules over the trunk about two weeks after a sore throat. Which condition and workup are most appropriate?

- A. Pityriasis rosea; reassurance with no testing
- B. Guttate psoriasis; obtain streptococcal throat culture and titers
- C. Nummular eczema; perform patch testing
- D. Secondary syphilis; obtain fungal cultures

6. A child with acute gastroenteritis and mild-to-moderate dehydration is able to drink. Compared with intravenous therapy, which statement about oral rehydration is most accurate?

- A. It is contraindicated whenever the sodium is abnormal
- B. It is effective first-line therapy, even for many children with hyponatremic or hypernatremic dehydration
- C. It becomes useful only after a trial of intravenous fluids has failed
- D. It is generally unable to replace the ongoing stool losses that continue during treatment

7. A 3-month-old former 28-week preterm infant requires inguinal hernia repair. Which postoperative complication most warrants apnea monitoring?

- A. Malignant hyperthermia
- B. Postoperative nausea and vomiting
- C. Emergence delirium
- D. Postoperative apnea and bradycardia

8. A 6-month-old infant is found to have a perineal vascular lesion noted since shortly after birth, accompanied by minor bleeding. Beyond addressing the lesion itself, which of the following assessments is most important to perform?

- A. Measurement of serum estradiol and luteinizing hormone levels
- B. A neurologic assessment for associated spinal dysraphism
- C. A forensic interview of the family for suspected abuse
- D. Vaginoscopy under anesthesia to exclude a foreign body

9. A neonate with Down syndrome has bilious vomiting and a double-bubble sign on abdominal radiograph. Which associated anomaly is most likely?

- A. Pyloric stenosis
- B. Duodenal atresia
- C. Biliary atresia
- D. Malrotation without volvulus

10. Immediately after endotracheal intubation, which step is mandatory to confirm that the tube is in the trachea rather than the esophagus?

- A. A single auscultation over the stomach only
- B. Observing the oxygen saturation alone for one minute
- C. Measuring the cuff pressure
- D. Detection of exhaled carbon dioxide by colorimetry or capnography

11. There is a well-recognized bidirectional association between Tourette disorder and which condition?

- A. Cystic fibrosis
- B. Type 1 diabetes mellitus
- C. Obsessive-compulsive disorder
- D. Chronic iron-deficiency anemia

12. Which set of effects is most consistently associated with exposure to media violence?

- A. A proven, direct causal link to later violent criminal behavior
- B. Increased aggression, desensitization, and reduced empathy in viewers
- C. A reliable reduction in aggressive thoughts and behaviors
- D. No measurable effect on arousal, empathy, or aggression at all

13. Under the National Institutes of Health criteria, the diagnosis of polycystic ovary syndrome requires which of the following combinations?

- A. Polycystic ovaries on ultrasonography plus hyperandrogenism
- B. Oligoovulation or anovulation plus hyperandrogenism
- C. Polycystic ovaries on ultrasonography plus oligoovulation
- D. Elevated luteinizing hormone-to-follicle-stimulating hormone ratio plus hyperandrogenism

14. Colostrum, the first milk produced after birth, is especially rich in which protective component?

- A. Milk sugar (lactose)
- B. Secretory immunoglobulin A
- C. Long-chain triglyceride fat
- D. The protein casein

15. In a child with gross hematuria and a normal coagulation profile, which microscopic finding best confirms an intrinsic glomerular cause?

- A. Dysmorphic red cells accompanied by red blood cell casts
- B. Numerous uniform red cells without casts
- C. Abundant squamous epithelial cells
- D. Calcium oxalate crystals with normal red cells

16. A 10-month-old has iron-deficiency anemia and intermittent blood-streaked stools after being given large amounts of cow's milk. Which mechanism most likely contributes to the iron deficiency?

- A. Autoimmune hemolysis of red cells
- B. Intestinal hookworm infection with blood loss
- C. Cow's-milk-protein-induced enteric blood loss
- D. Splenic sequestration of red cells

17. Köhler disease refers to avascular necrosis of which bone?

- A. The tarsal navicular
- B. The second metatarsal head
- C. The calcaneal apophysis
- D. The base of the fifth metatarsal

18. What is the principal physiologic role of carnitine in fatty acid metabolism?

- A. Transporting long-chain fatty acids into mitochondria
- B. Activating intestinal absorption of dietary fat
- C. Synthesizing cholesterol from acetyl-CoA
- D. Storing triglyceride within adipose tissue

19. A newborn has a midline abdominal wall defect at the umbilical ring in which herniated viscera are covered by a membranous sac. Compared with gastroschisis, this lesion is most notable for which of the following?

- A. Complete absence of any covering membrane over the bowel
- B. A location consistently to the right of the umbilical cord
- C. An almost exclusive occurrence as an isolated finding
- D. A strong association with chromosomal and other major anomalies

20. A 10-year-old from the south-central United States presents in July with five days of fever, headache, and myalgia after camping; he reports a tick bite. He has a faint maculopapular rash, mild splenomegaly, and altered mental status. Laboratory testing shows leukopenia, thrombocytopenia, elevated aminotransferases, and hyponatremia. Which infection best fits this presentation?

- A. Acute Epstein-Barr virus infectious mononucleosis
- B. Disseminated meningococcemia with early septic shock
- C. Acute lymphoblastic leukemia presenting with cytopenias
- D. Human monocytic ehrlichiosis from *Ehrlichia chaffeensis*

21. A child with progressive sensorineural loss has an enlarged vestibular aqueduct on CT and a positive perchlorate discharge test. Which gene is most likely involved?

- A. GJB2 (connexin-26)
- B. PAX3
- C. MYO7A in isolation
- D. SLC26A4 (pendrin)

22. A 15-year-old girl has 6 days of more than five bloody stools daily, fever, tachycardia, severe anemia, hypoalbuminemia, and leukocytosis; abdominal radiograph shows a dilated transverse colon. Which intervention is contraindicated in this setting?

- A. Intravenous corticosteroids
- B. Barium enema
- C. Stool studies for enteric pathogens
- D. Intravenous fluid and electrolyte repletion

23. A former 25-week infant with bronchopulmonary dysplasia has persistent tachypnea, increased work of breathing during feeds, and poor weight gain despite adequate calories. Which of the following is the most appropriate management priority?

- A. Optimize nutrition and energy intake for growth
- B. Restrict total calories to lower metabolic demand
- C. Begin chronic daily oral corticosteroid therapy
- D. Stop supplemental oxygen to stimulate breathing

24. A neonate has severe symptomatic hypercalcemia, while an asymptomatic parent has lifelong mild hypercalcemia with low urine calcium. Which disorder links these findings?

- A. Vitamin D intoxication in both the neonate and the parent
- B. Primary hyperparathyroidism from a parathyroid adenoma
- C. Dietary calcium excess in the breastfeeding pair
- D. A calcium-sensing receptor defect, severe when biallelic and benign when monoallelic

25. Which is characteristic of the familial pattern of dyslexia?

- A. Approximately 50% of children who have a parent with dyslexia will also have dyslexia
- B. Dyslexia is inherited in an autosomal dominant pattern linked to a single identifiable gene
- C. Risk in siblings of affected individuals is approximately 10%, reflecting low heritability
- D. Familial occurrence is limited to paternal transmission and does not follow maternal lines

26. To best reduce coronary artery complications, the diagnosis and treatment of Kawasaki disease should ideally occur within which window?

- A. Within 10 days, ideally within 7 days
- B. Within 30 days of the initial fever onset
- C. After the fever has fully resolved
- D. Once coronary aneurysms have appeared

27. Calcium EDTA chelation requires monitoring for which characteristic toxicity?

- A. Progressive irreversible sensorineural hearing loss
- B. Acute hemorrhagic cystitis with passage of clots
- C. Immediate anaphylaxis upon the very first dose
- D. Nephrotoxicity with proteinuria and rising creatinine

28. Which abnormality underlies the most common T-cell immunodeficiency disorder?

- A. Mutation of the AIRE gene
- B. Mutation of the WAS gene
- C. A chromosome 22q11.2 deletion
- D. A defect in ZAP-70 signaling

29. A family requests therapy to change their adolescent's same-sex orientation. Which response reflects the position of major medical organizations?

- A. A gay or lesbian orientation is not an illness and need not be changed
- B. Brief behavioral therapy reliably converts orientation to heterosexual when started early enough
- C. Orientation-change counseling is appropriate whenever a family strongly prefers that approach
- D. A same-sex orientation is a psychiatric disorder that warrants corrective treatment

30. Which mediator drives the iron sequestration of anemia of chronic disease by promoting internalization of ferroportin?

- A. Erythropoietin
- B. Ferritin
- C. Hepcidin
- D. Transferrin

31. In a neonate or young infant aged 0-3 months, what minimum rectal temperature defines fever without a focus?

- A. 37.5C (99.5F)
- B. 38.5C (101.3F)
- C. 38.0C (100.4F)
- D. 39.0C (102.2F)

32. According to symptom-based pediatric criteria, infant colic requires recurrent crying or fussing without obvious cause occurring in an infant of which age range?

- A. Younger than 2 months
- B. Between 6 and 12 months
- C. Between 12 and 18 months
- D. Younger than 5 months

33. Which medication is commonly used to lower cerebrospinal fluid production in idiopathic intracranial hypertension?

- A. Oral prednisone
- B. Indomethacin
- C. Sumatriptan
- D. Acetazolamide

34. A 7-year-old girl has mild proximal weakness and a creatine kinase value of a few thousand units; her brother has a confirmed dystrophinopathy; which mechanism best explains her symptoms?

- A. Skewed X-inactivation per the Lyon hypothesis
- B. A new autosomal dominant point mutation
- C. Mitochondrial heteroplasmy in muscle
- D. Homozygous dystrophin gene deletion

35. A survivor of Hodgkin lymphoma treated with chest radiation and an anthracycline asks about long-term risks. Which combination is most important to monitor?

- A. Hearing loss and chronic nephropathy
- B. Breast cancer and cardiac dysfunction
- C. Hemorrhagic cystitis and male infertility
- D. Pulmonary fibrosis and posterior cataracts

36. A variant at the boundary between an intron and an exon disrupts normal splicing. Which consequence is expected?

- A. Addition of a longer poly-A tail
- B. Substitution of a single amino acid only
- C. Exon skipping or retention of intronic sequence
- D. Increased rate of gene transcription

37. Which statement about physical growth in middle childhood is MOST accurate?

- A. Growth is perfectly steady and continuous day to day
- B. Growth is discontinuous, occurring in several irregular spurts each year
- C. Head circumference increases by about 10 cm over the period
- D. Linear growth ceases entirely until puberty

38. A teenager who wears soft contact lenses develops itching and large papillae on the upper tarsal conjunctiva. Which condition is most likely?

- A. Vernal keratoconjunctivitis
- B. Seasonal allergic conjunctivitis
- C. Bacterial conjunctivitis
- D. Giant papillary conjunctivitis

39. A child in ventricular fibrillation has received one shock at 2 J/kg followed by 2 minutes of CPR, and the rhythm remains ventricular fibrillation. After delivering a second shock, what is the recommended next action?

- A. Check a pulse for a full 60 seconds before resuming compressions
- B. Administer sodium bicarbonate before resuming compressions
- C. Immediately resume chest compressions and give epinephrine 0.01 mg/kg
- D. Withhold compressions and obtain a 12-lead electrocardiogram

40. A sinus venosus atrial septal defect is most often associated with which additional anomaly?

- A. A discrete subaortic membrane causing obstruction
- B. Apical displacement of the tricuspid leaflets
- C. Partial anomalous pulmonary venous connection of a right vein
- D. Supravalvar narrowing of the ascending aorta

41. A 15-year-old girl with mild intellectual disability (ID) who previously received services under an individualized education program (IEP) is transitioning to a community college vocational program. Her parents are concerned about her future independence. According to outcomes data, what is the approximate employment rate for adults with ID?

- A. Only about 15% of adults with ID are estimated to have jobs
- B. Employment rates for adults with ID approximate those of the general population
- C. About 75% of adults with mild ID are employed in supported employment settings
- D. Approximately 50% of adults with ID are employed in competitive positions

42. In mass psychogenic illness, symptoms most often spread through which mechanism?

- A. Direct bloodborne transmission passing between individuals
- B. Ingestion of a single common contaminated food
- C. The bite of a single shared arthropod vector
- D. Seeing or hearing others who are affected

43. An 8-year-old boy has intense ocular itching, photophobia, ropy discharge, and giant cobblestone papillae on the upper tarsal conjunctiva that worsen each spring. Which is the most likely diagnosis?

- A. Seasonal allergic conjunctivitis
- B. Giant papillary conjunctivitis from contact lenses
- C. Vernal keratoconjunctivitis
- D. Bacterial conjunctivitis

44. An injury is classified as ASIA Impairment Scale grade A when which condition is met?

- A. No sensory or motor function is preserved in sacral segments S4-S5
- B. Some motor function is preserved more than three levels below the lesion
- C. Light touch is preserved at S4-S5 but motor function is absent below
- D. At least half of key muscles below the lesion are graded 3 or better

45. Renal clearance of gentamicin is lowest, requiring the longest dosing interval, in which group?

- A. Children aged two to five years
- B. Adolescents aged twelve to eighteen years
- C. Neonates in the first weeks of life
- D. School-age children aged six to ten years

46. Asymptomatic bacteriuria in an otherwise healthy child is best managed in which way?

- A. Generally without antibiotic treatment
- B. With a prolonged antibiotic course
- C. With prompt surgical evaluation
- D. With chronic catheter drainage

47. Bilateral itching, watery discharge, and conjunctival chemosis that recurs seasonally most characterize which type of conjunctivitis?

- A. Gonococcal conjunctivitis
- B. Bacterial purulent conjunctivitis
- C. Chlamydial conjunctivitis
- D. Allergic conjunctivitis

48. Which is the treatment of choice for isolated moderate-to-severe valvar pulmonary stenosis?

- A. Long-term oral beta-blocker therapy
- B. Percutaneous balloon valvuloplasty
- C. Continuous prostaglandin E1 infusion
- D. Surgical pulmonary artery banding

49. Which feature most strongly favors spontaneous intestinal perforation over necrotizing enterocolitis in a preterm infant?

- A. Extensive pneumatosis intestinalis preceding the perforation
- B. An isolated perforation early in life without preceding pneumatosis
- C. Onset only after full enteral feeds with portal venous gas
- D. Bloody stools accompanied by widespread intramural gas

50. Epipodophyllotoxins such as etoposide are notably associated with which delayed complication?

- A. Delayed hemorrhagic cystitis
- B. Late dilated cardiomyopathy
- C. Progressive pulmonary fibrosis
- D. Secondary acute myeloid leukemia

Answer sheet

Name: _____ Exam code: 143YVH2

Circle one letter per question.

1. A B C D

2. A B C D

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